ModelIt!

Student Activity Pack

Why Are Cheetahs Losing Their Speed?

Genetic Diversity, Habitat Loss, and the Future of the Fastest Animal

Grade 07L2 | NGSS MS-LS4-6, MS-LS4-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think there are different breeds of dogs with very different traits, but all cheetahs look almost identical?**

2. **What do you think happens to a population when there are only a few individuals left?**

3. **Draw what you think a healthy population's genetic diversity looks like compared to an endangered species.**

4. **How do you think roads and cities affect wild animal populations?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Population Size | Habitat Fragmentation | Genetic Diversity | Trait Variation | Species Fitness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Severe Bottleneck

Very small population, highly fragmented habitat

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Healthy Population

Large population, connected habitat

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Conservation Corridor

Gradually increasing population, decreasing fragmentation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Disease Susceptibility — How vulnerable the population is to epidemics, which increases when all individuals share similar immune genes

Inbreeding Rate — The frequency of mating between closely related individuals, which increases in small, isolated populations

Gene Flow — The transfer of genetic material between separate populations through migration, which introduces new genetic variation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Cheetahs are the fastest land animals on Earth, but they're slowing down. How can an entire species lose the very trait that defines its survival? Use evidence from your simulation to support your answers.

1. **Why is genetic diversity sometimes called the 'insurance policy' of a species? How does your model support this idea?**

2. **How does the cheetah's genetic bottleneck 10,000 years ago still affect the species today?**

3. **If all cheetahs are genetically similar, why is that a problem for fighting diseases?**

4. **How does habitat fragmentation make the genetic diversity problem worse, even if the total population stays the same?**

5. **What would your model predict for human-managed breeding programs that intentionally mix individuals from different isolated populations?**

8 STEM Challenge: Design a Cheetah Conservation Plan

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YOUR ENGINEERING MISSION

Create a science-based conservation plan to restore genetic diversity in cheetah populations, using your simulation model data to justify your strategies.

THE SCENARIO

The Cheetah Conservation Fund has asked your team to develop a genetic rescue plan. Only about 7,000 cheetahs remain in the wild, scattered across fragmented habitats in Africa and Iran. Your plan must address both habitat connectivity and genetic diversity using evidence from your model.

Engineering Design Thinking

1. WHAT GEOGRAPHIC REGIONS WILL YOUR CONSERVATION PLAN TARGET?
2. HOW WILL YOU RECONNECT FRAGMENTED HABITATS TO ALLOW GENE FLOW?
3. WHAT MODEL DATA WILL YOU USE TO JUSTIFY YOUR MINIMUM POPULATION SIZE RECOMMENDATIONS?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN IS SUCCESSFULLY INCREASING GENETIC DIVERSITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation geneticists study the genetic health of endangered species and design breeding and habitat programs to preserve genetic diversity. They work at wildlife agencies, zoos, universities, and

conservation organizations, earning $55,000-$110,000/year.